

The State of Hybrid and Electric Vehicle Adoption in the U.S.

Sean Bush

Hybrid and Electric Vehicles are a growing sector of the automobile market in the U.S. While gas-powered vehicles continue to make up the majority of the market, these gas alternatives are making up a larger percentage of the market each year due to their superior fuel efficiency and environmental friendliness, and due to increasing infrastructure support and manufacturer adoption. It is not unreasonable to assume that in the near future, gas-alternative cars will comprise an even larger portion of the market. This report seeks to assess the current state of Hybrid and Electric Vehicle (H/EV) adoption in the United States in 2026 by visualizing data concerning H/EV sales in Tableau, combining them into several dashboards comprising a single story, and publishing that story to the Tableau Public web platform.

Research Questions

This report seeks to provide a general overview of the state of H/EV adoption in the U.S. and, as such, addresses several different research questions rather than one singular one. These questions include:

- How have U.S. H/EV Sales been growing over the past several years?
- In the U.S., are emissions being reduced in concert with growing H/EV sales?
- How does H/EV adoption vary across states?
- How is adoption predicted to change over the next year?

As will be seen in the following two sections, these questions were used to guide the data gathering and data analysis process in order to ensure that the gathered data and the analysis stay relevant to the goals of the report.

Data Gathering

Four data sets were assembled to address each of the four research questions, each pulled from open government data or published with a license allowing for free use of the data. The first of these datasets is titled Light Duty Electric Drive Vehicles Monthly Sales Updates and was published by Argonne National Laboratory (n.d.). This dataset records monthly sales numbers from as early as December 2010 all the way to May 2026 (at the time this data was retrieved for the project). Sales are split into four categories: standard Battery Electric Vehicles (BEV), Hybrid Electric Vehicles (HEV), Plug-in Hybrid Electric Vehicles (PHEV), and total Light-Duty Vehicles (LDV). Light-duty gas vehicle sales can be estimated by subtracting all others from LDV sales. This dataset was used to address the research question “How have U.S. H/EV Sales been growing over the past several years?” It was also used to address the question “How is adoption predicted to change over the next year?” through Tableau’s prediction capabilities.

The second dataset is the Greenhouse Gas Emissions dataset, published by the Environmental Protection Agency in their report “Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990-2022 U.S.” (EPA, 2024). This dataset records estimates of yearly net greenhouse emissions in the U.S. from 1990 to 2022. The dataset records estimates for many different kinds of greenhouse gases, though this analysis used only the net total greenhouse gas emissions figures for each year. Also, the written report published by the EPA only lists select years, but the full dataset was published alongside it with figures for all years between 1990 and 2022. This dataset was used to address the question “In the U.S., are emissions being reduced in concert with growing H/EV sales?” This dataset is appropriate for this use case because it provides emissions data that can be compared to the sales data from the previous dataset.

The third dataset is the 2024 Light-Duty Vehicle Registration Counts by State and Fuel Type dataset, published by the Department of Energy (n.d.). The dataset lists the number of registered vehicles in each state for various fuel types including gas, hybrid, and electric. This set was used to address the question “How does H/EV adoption vary across states?”

The fourth dataset is the 2024 National State Population Estimates published by the U.S. Census Bureau (2024). This dataset records estimated populations for each state in the U.S. This set was used alongside the previous to answer the question “How does H/EV adoption vary across states?” This set allowed for population-adjusted comparisons to be made between states to further clarify their differences.

One more data source was used for this analysis, but rather than being gathered from a published dataset, it is instead sourced from the All Stations web API hosted by the National Laboratory of the Rockies on the Developer Network branch of their website (n.d.). This API provides real-time information on all electric vehicle charging stations in the U.S., including their location and specifications. This set was also used to address the question “How does H/EV adoption vary across states?” by comparing the number of EV charging stations across states.

Since the four datasets were already published as tables, all were relatively easy to prepare by either directly downloading as a CSV file or by copying into a CSV file. The API data source was significantly more difficult to prepare, as it was desirable to utilize the real-time nature of the API so that the data could be regularly updated.

Data Preparation

The process of getting the Tableau worksheet to have regularly updated data from the NLR API required several steps. Tableau only allows a few kinds of regularly updated data connections on the Tableau Public platform. One of those options is through a connection to a Google Sheet. In Tableau, you have the option of connecting Tableau to a Google account where it can retrieve data from various Google Sheets. Once uploaded to the public platform, it can then refresh that data by pulling from the Google Sheet once a day.

With that connection made, the other requirement for this setup to function is that the Google Sheet gets regularly updated data from the API. This was accomplished in this project through the use of a Python script run regularly by GitHub Actions. The code responsible for this step can be found in this project's [GitHub Repository](#). The script responsible for the code is `update_data.py`, which connects to the NLR API using an API Key provided by their website and stored in an encrypted state with GitHub Secrets. It retrieves the data for all stations from the API and counts the number of stations.

```
def retrieve_fuel_station_data(api_key: str,
                               request_url: str = 'https://developer.nlr.gov/api/alt-fuel-stations/v1.json'
                               ) -> dict[str, int]:

    request_parameters = {
        'api_key': api_key,
        'status': 'E,T',
        'access': 'public',
        'fuel_type': 'ELEC'
    }

    response = requests.get(request_url, request_parameters).json()

    stations = response['fuel_stations']

    station_counts_per_state = {}

    for station in stations:
        state = station['state']
        if state not in station_counts_per_state:
            station_counts_per_state[state] = 0
        station_counts_per_state[state] += 1

    return station_counts_per_state
```

The program then authenticates with the Google Sheets API using credentials also stored in GitHub secrets and uploads the data to the Google Sheet.

```
def update_google_sheet(station_counts_per_state: dict[str, int],
                        sheet_id: str,
                        credentials: dict[str, Any]) -> None:

    scopes = ['https://www.googleapis.com/auth/spreadsheets']
    client = gspread.service_account_from_dict(credentials)

    sheet = client.open_by_key(sheet_id)

    states = sheet.sheet1.col_values(1)

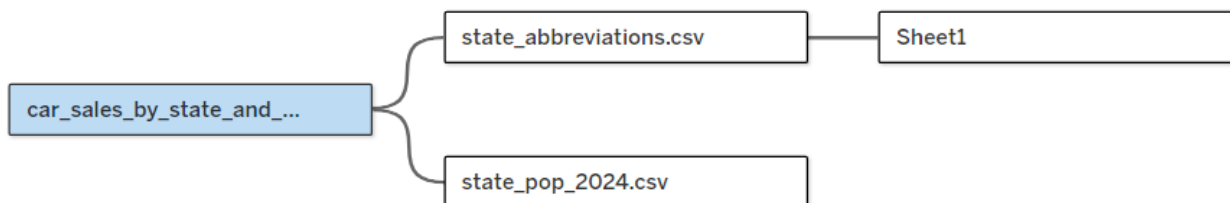
    for row, state in enumerate(states):
        count = station_counts_per_state[str(state)] if str(state) in station_counts_per_state else 0
        sheet.sheet1.update_cell(row + 1, 2, count)

    for s, c in zip(sheet.sheet1.col_values(1), sheet.sheet1.col_values(2)):
        print(f'{s}: {c}')
```

With the connection between the API and the workbook established and the four other datasets stored as CSV files on my system, the final step in the data preparation process was to define the relationships between the tables in Tableau. The tables were divided into two different data sources. The first combined the Monthly Light Vehicle Sales data with the EPA Yearly Emissions data. The relationship was formed by connecting the year associated with each month-year row in the Monthly Sales data and the year column of the EPA data.



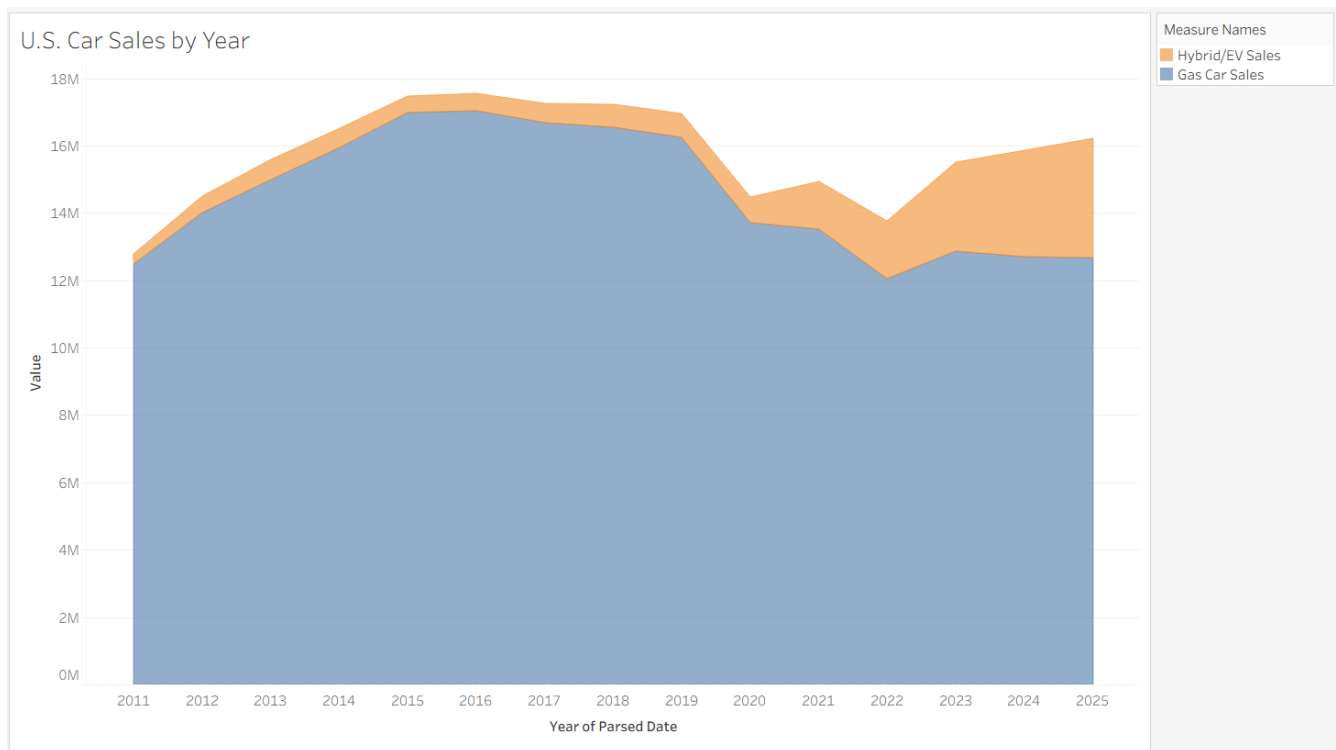
The second data source combined the per-state car registrations data with the state population data and the NLR data retrieved from the Google Sheet. The car registrations and state populations tables were connected with a relationship between their state columns. The car sales data was also connected to the NLR data on their state columns, but another dataset had to be created called `state_abbreviations.csv` to connect each state's name with its abbreviation since the former is used in the registration dataset and the latter is used in the NLR dataset.



Analysis

The data was then used to create several different visualizations, which were turned into three dashboards. Those dashboards were then connected into a single story to highlight the key takeaways from the analysis. The completed data story can be found uploaded to the Tableau Public platform as a public workbook entitled [The State of Hybrid/Electric Vehicle Adoption in the U.S.](#) This section of the report will go through each of the visualizations, briefly explaining how they were made and how they contributed to answering the research questions.

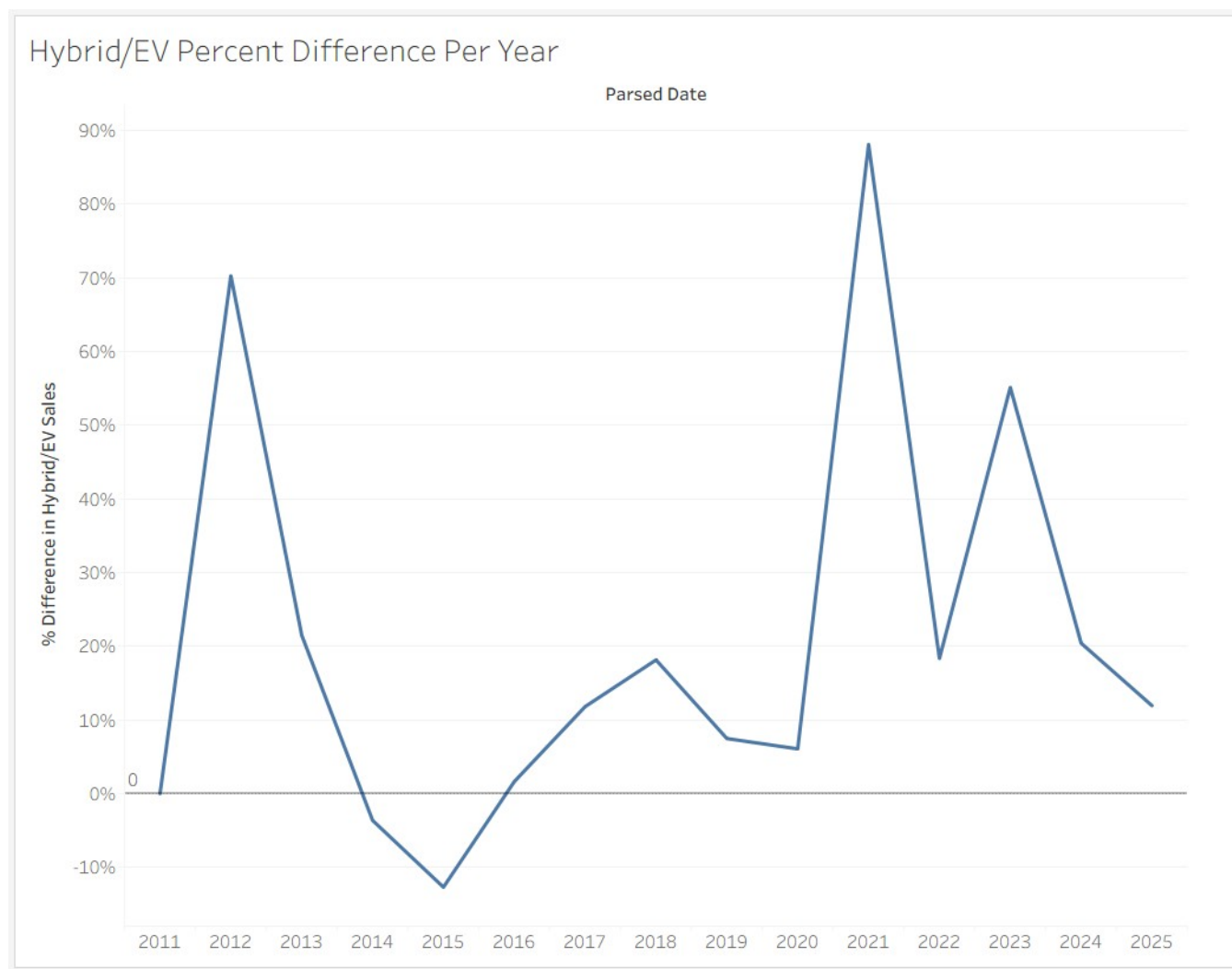
The first dashboard is titled The State of Hybrid/EV Sales in the U.S. and is comprised of three visualizations. The first is a line/area plot of Light Vehicle Sales in the U.S. from 2011 to 2025. The rationale for the inclusion of this plot is that it addresses the research question “How have U.S. H/EV Sales been growing over the past several years?” by comparing H/EV sales to Gas car sales over the past several years. The data used for this visualization is the Light Duty Monthly Sales dataset. The x-axis uses the years parsed from the month-year column of that dataset. The y-axis shows the sum of all sales that year, divided into H/EV and Gas car sales. Colors are used to distinguish the two, with the area in blue representing gas car sales and the orange area representing H/EV sales.



The visualization shows two main crucial pieces of information. The first is the general trend of car sales over the past fifteen years. Sales were generally trending upward until 2016. Then they declined slightly until 2020, when the COVID-19 pandemic began, which spurred a decline over the next several years, followed by a rebound from 2023 to the present. The second piece of information is that in the years since the pandemic, the proportion of H/EV

sales has grown significantly such that as of 2025, it comprises a significant portion of the market.

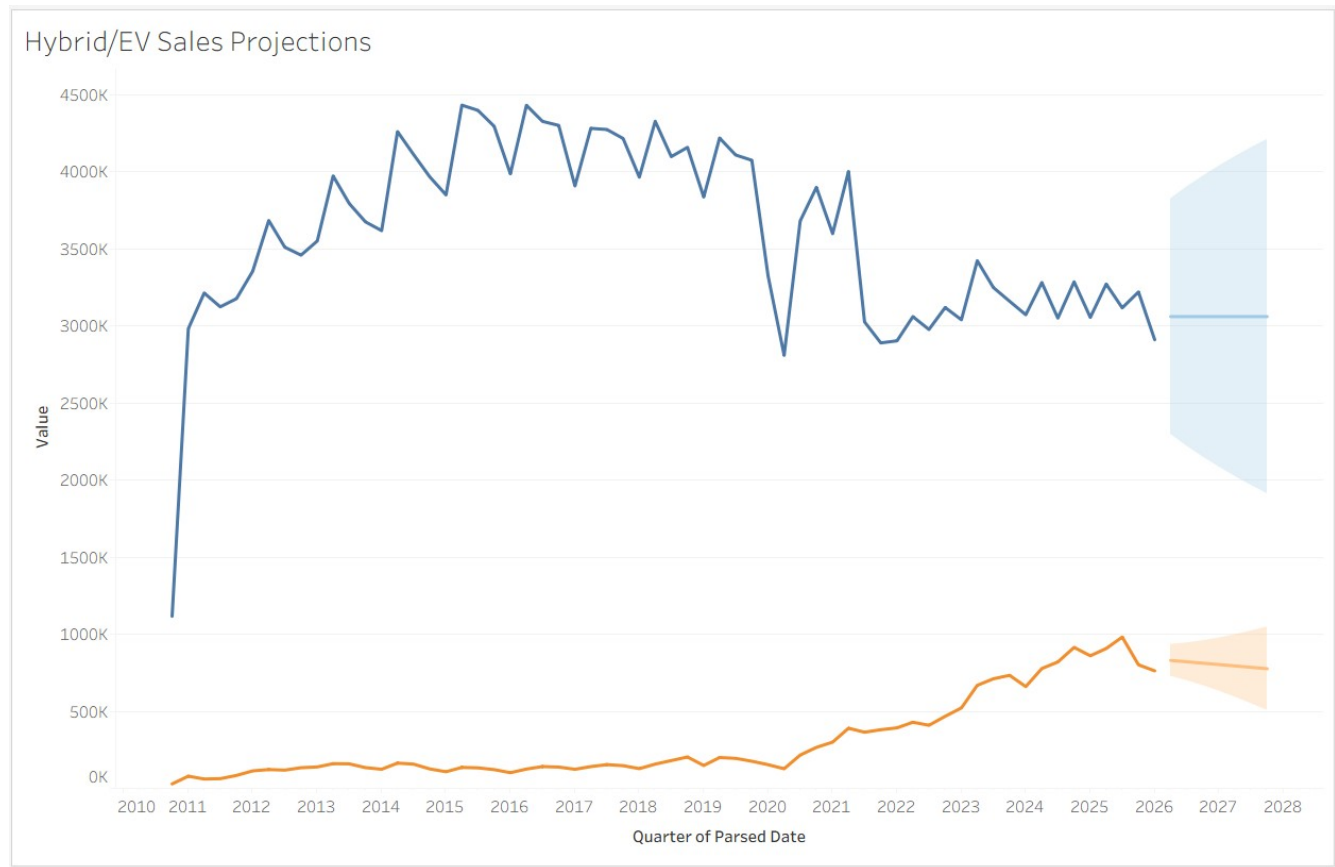
The second visualization is a line plot of the percent difference of Hybrid/EV sales from 2011 to 2025. The plot addresses the research question “How have U.S. H/EV Sales been growing over the past several years?” The rationale for this plot’s inclusion is that it addresses an aspect of that research question that could be somewhat obscured in the previous analysis: which years specifically have sales increased or decreased and by how much? The data for this plot was sourced from the Light Duty Monthly Sales dataset. The x-axis is the years as parsed from that dataset’s month-year column, and the y-axis is the percent difference of each year’s total H/EV sales. Also included is a reference line at 0%. Any data point above this line shows an increase in sales from the previous year, and any point below shows a decrease in sales from the previous year.



The plot shows that Hybrid/EV sales have been very consistent in increasing year over year, only ever decreasing in 2015 and 2016. However, the amount that they increase by is

somewhat erratic. For example, 2012, 2021, and 2023 saw very large gains, whereas 2022 and 2025 saw fairly small gains.

The third visualization is a line plot where gas and H/EV sales are plotted separately (as opposed to in visualization one, where they were stacked on top of one another), and sales are predicted for the following quarters all the way through Q4 2027. This addresses the research question “How is adoption predicted to change over the next year?” by projecting possible sales values for the next year. This visualization also used the Light Duty Monthly Sales dataset. The x-axis in this plot is the quarters parsed from the year-month column of that dataset, and the y-axis is the quarterly sales figures for the two types of vehicles.

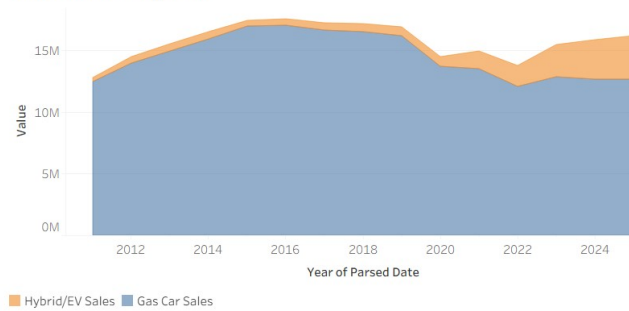


Tableau’s forecast model was used to project sales data into the future. The model predicts constant sales of gas vehicles, albeit with a large margin of error, allowing for unexpected significant gains or losses. The model predicts a slight decline in H/EV sales, probably spurred on by the decline in sales from Q3 2025 to Q1 2026. Its margin of error does allow for an increase or decrease, however, though the range is much smaller than the range for the gas vehicles margin of error. Even if sales follow the predictions closely, however, the predicted changes are not very significant. Overall, the model does not predict significant sales changes in 2027.

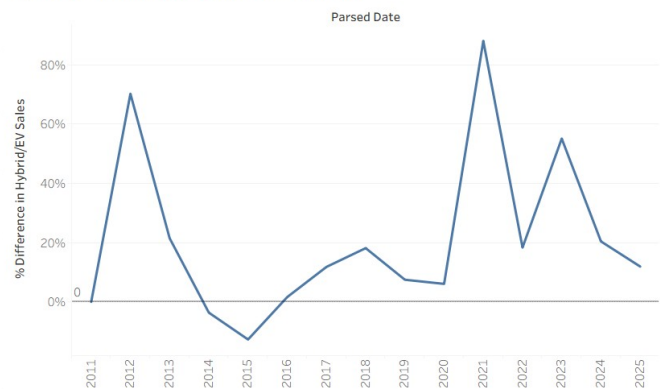
Those three visualizations comprise the first dashboard which appears as follows:

The State of Hybrid/EV Sales in the United States

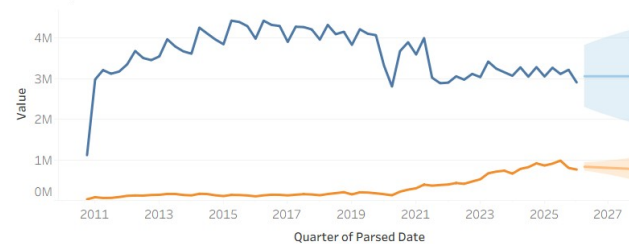
U.S. Car Sales by Year



Hybrid/EV Percent Difference Per Year



Hybrid/EV Sales Projections

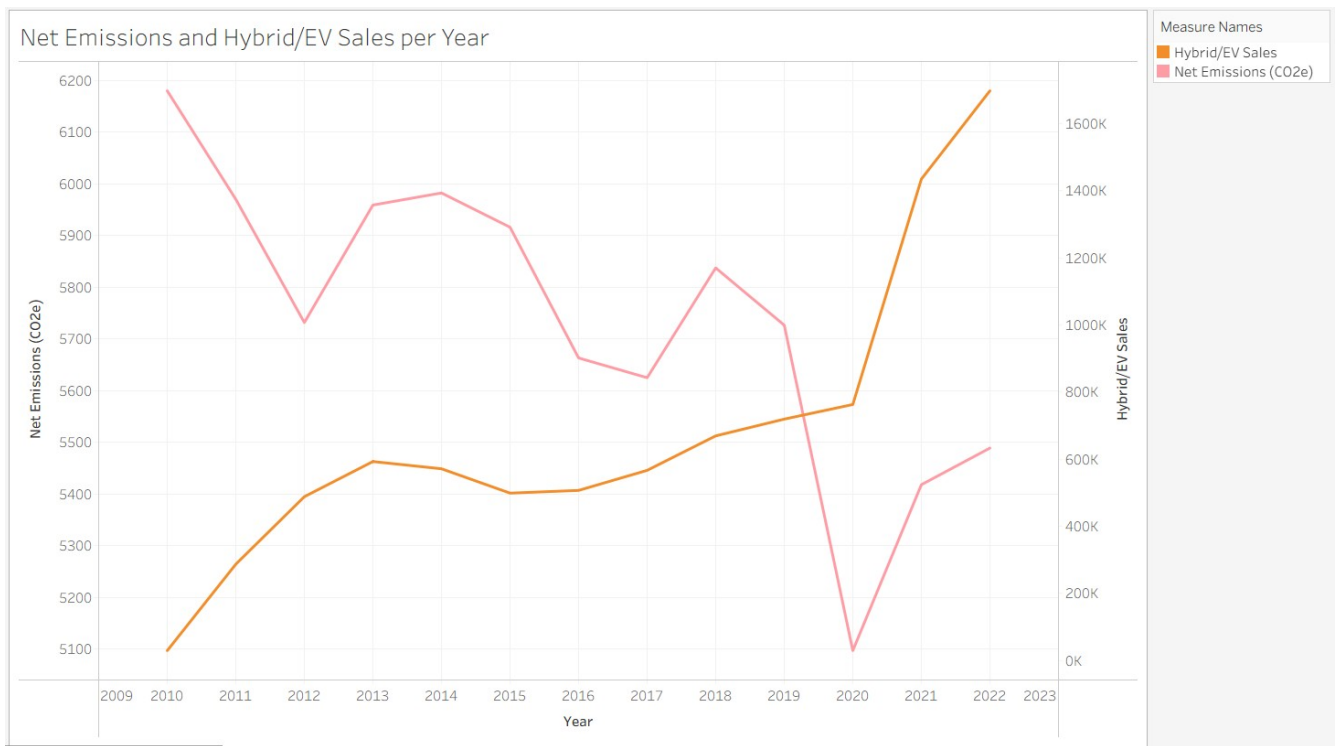


While still maintaining a minority share of the market, Hybrid and EV vehicle sales continue to increase every single year. However, the market is not projected to change much in 2027.

A summary statement is included at the bottom to state the overall takeaway from this dashboard: While still maintaining a minority share of the market, Hybrid and EV vehicle sales continue to increase every single year. However, the market is not projected to change much in 2027.

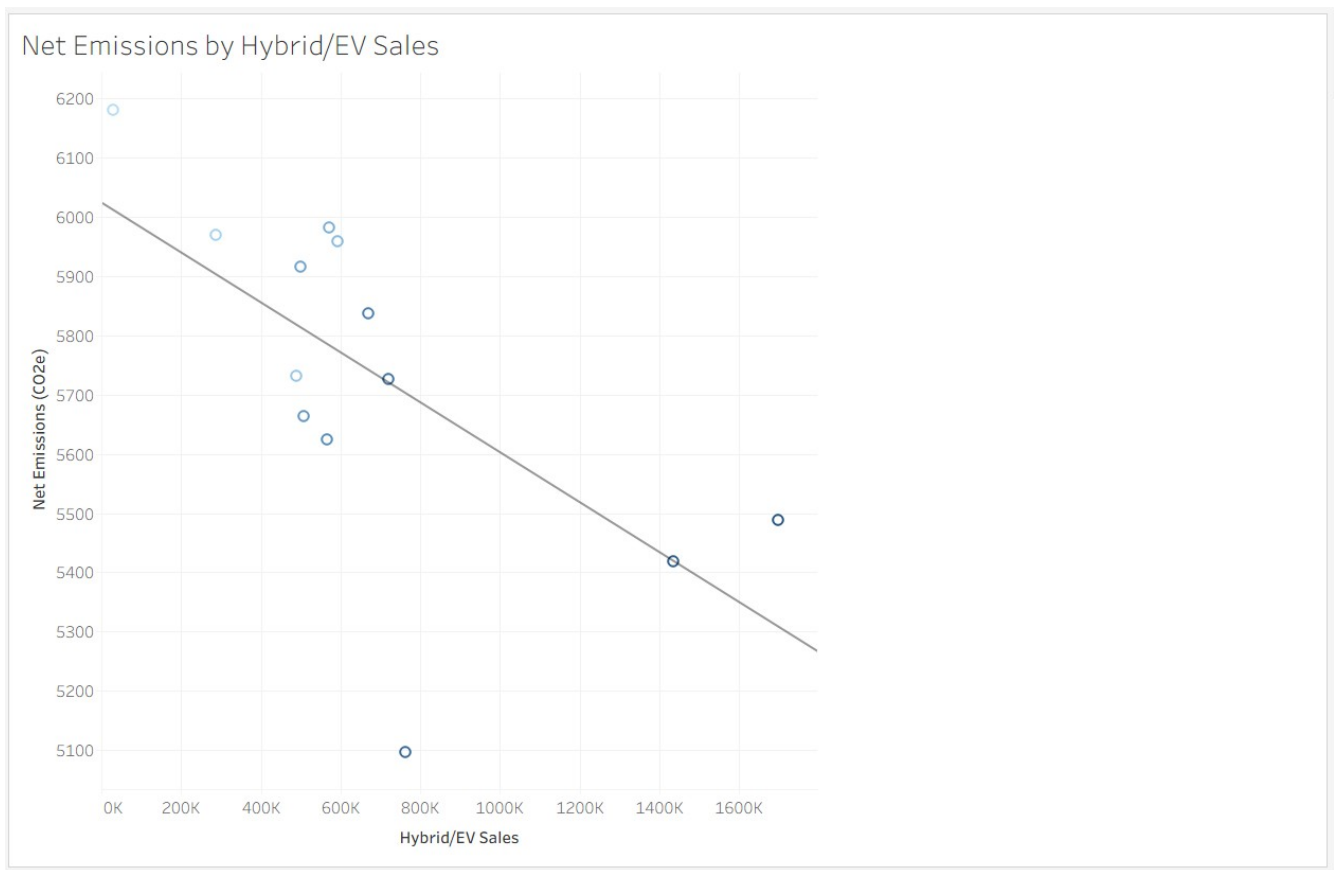
The second dashboard covers emissions data over the past several years to track if they trend alongside H/EV sales, since reducing emissions is a popular incentive for the manufacturing of H/EV cars. All visualizations in this dashboard address the research question “In the U.S., are emissions being reduced in concert with growing H/EV sales?”

The first visualization plots Net Greenhouse Gas Emissions in the U.S. against H/EV sales. Emissions data comes from the EPA dataset, and H/EV sales come from the Light Duty Vehicles Sales dataset. The visualization plots both emissions and sales as line graphs. The x-axis represents the year. The EPA data is already listed on a per-year basis; for the sales data, the year is parsed from the year-month column. These can both be plotted since they are related to each other in these columns, as discussed in the data preparation section of this report. This plot uses a dual y-axis since the lines are on different scales. The left y-axis corresponds with the gas emissions and represents the emissions (in millions of megatons of carbon dioxide equivalent) for a given year. The right y-axis represents the H/EV sales of a given year.



The graph shows that there is a general trend of emissions declining between 2010 and 2022, alongside a general increase in H/EV sales in those years.

The second visualization of this dashboard is a scatter plot showing the relationship between emissions and H/EV sales, regardless of time. There is also a trend line provided by Tableau's Analytics pane, which tries to show the general direction of the relationship. The data is sourced from the same datasets as the previous visualization. The x-axis represents H/EV Sales numbers and the y-axis represents net emissions.



The plot and the line of best fit show a general negative relationship between the two variables. As such, it can be concluded that these values move in opposite directions: when one increases, the other decreases.

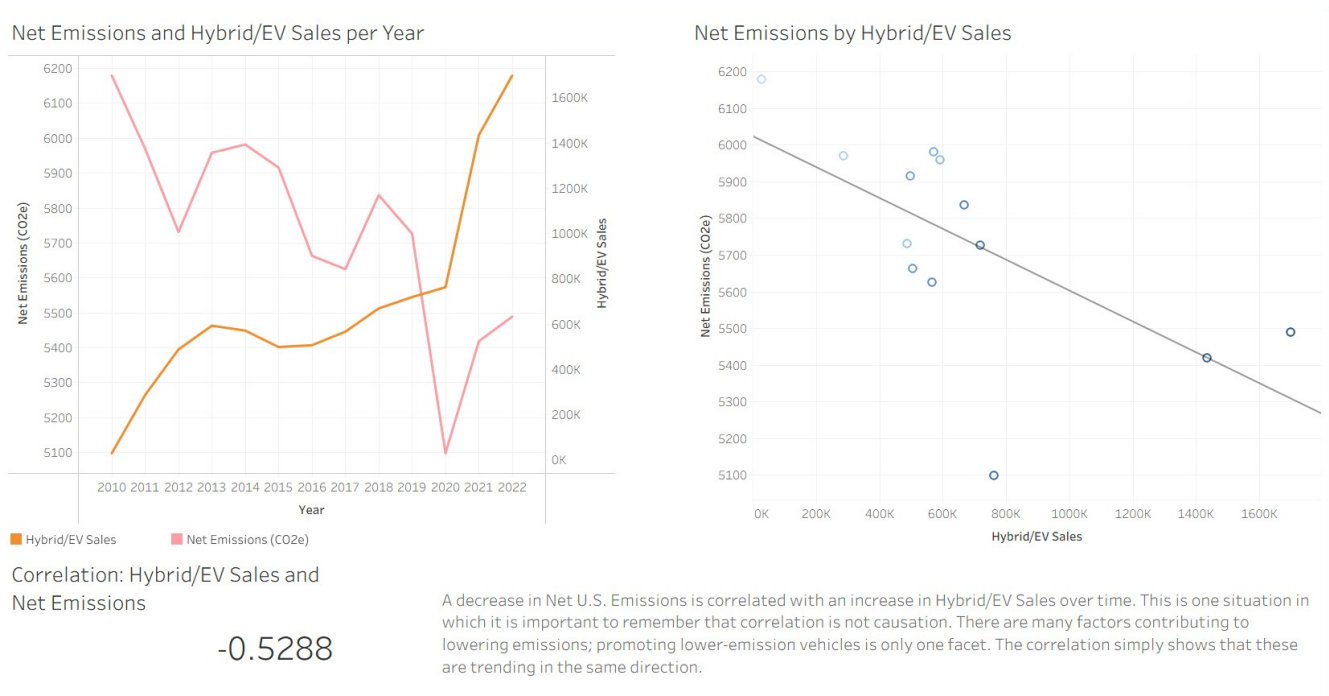
The third component of the dashboard is not a visualization, but simply a key metric: the Pearson coefficient of correlation between the variables. This metric quantifies the direction and strength of the relationship between emissions and H/EV sales.

Correlation: Hybrid/EV Sales and Net Emissions

-0.5288

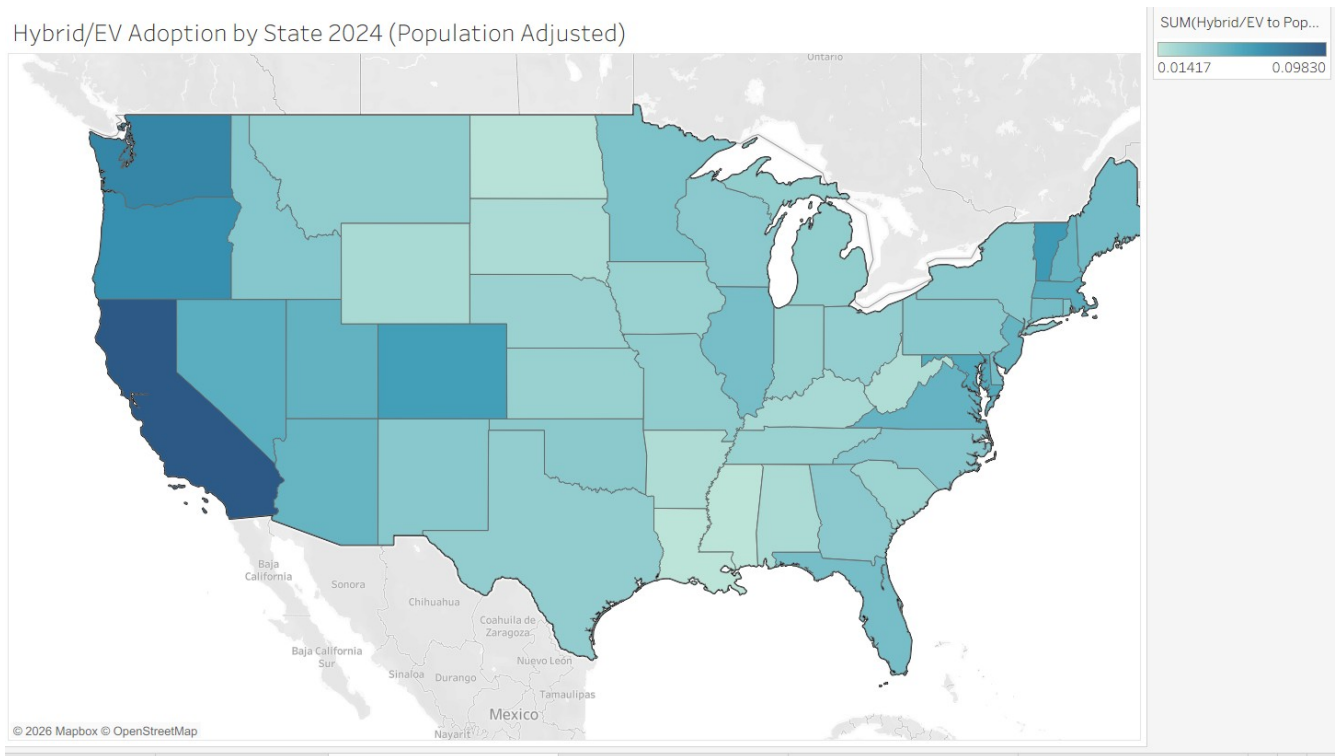
The correlation value is about -0.5, indicating a significant but far from perfect negative correlation. This corresponds well with the scatter plot, which showed a general negative relationship but had some data points quite far from the trend line. It is important to note, at this point, that this is an important example of correlation not equating to causation. Even if there were a causal relationship, the emissions data used for this analysis is an estimate of all U.S. emissions, and there are many different factors influencing emissions besides more environmentally friendly cars being on the roads. If one wanted to specifically establish a

causal link, it would require a robust model that controls for many more factors, and that is beyond the scope of this analysis. The research question was, “In the U.S., are emissions being reduced in concert with growing H/EV sales?” and the analysis suggests that the answer is yes. As emissions are declining, H/EV sales are also increasing. The combined dashboard says as much in its description.



The third and final dashboard addresses the research question “How does H/EV adoption vary across states?” The first visualization does so by showing a map of the U.S. with the color of each state corresponding to the proportion of the population owning a H/EV. The population data is sourced from the Census Bureau 2024 Population Estimates dataset, and the adoption figures are sourced from the Department of Energy’s 2024 Light-Duty Vehicle Registrations dataset. The plot uses Tableau’s ability to create maps out of geographic data. Each U.S. state gets its own location on the map. The color of each state corresponds with the ratio between that state’s H/EV sales and its population in 2024. This ensures that states can be compared to each other regardless of their population. A deeper blue corresponds with a larger fraction of the state possessing H/EV cars, while a light blue represents a lesser portion.

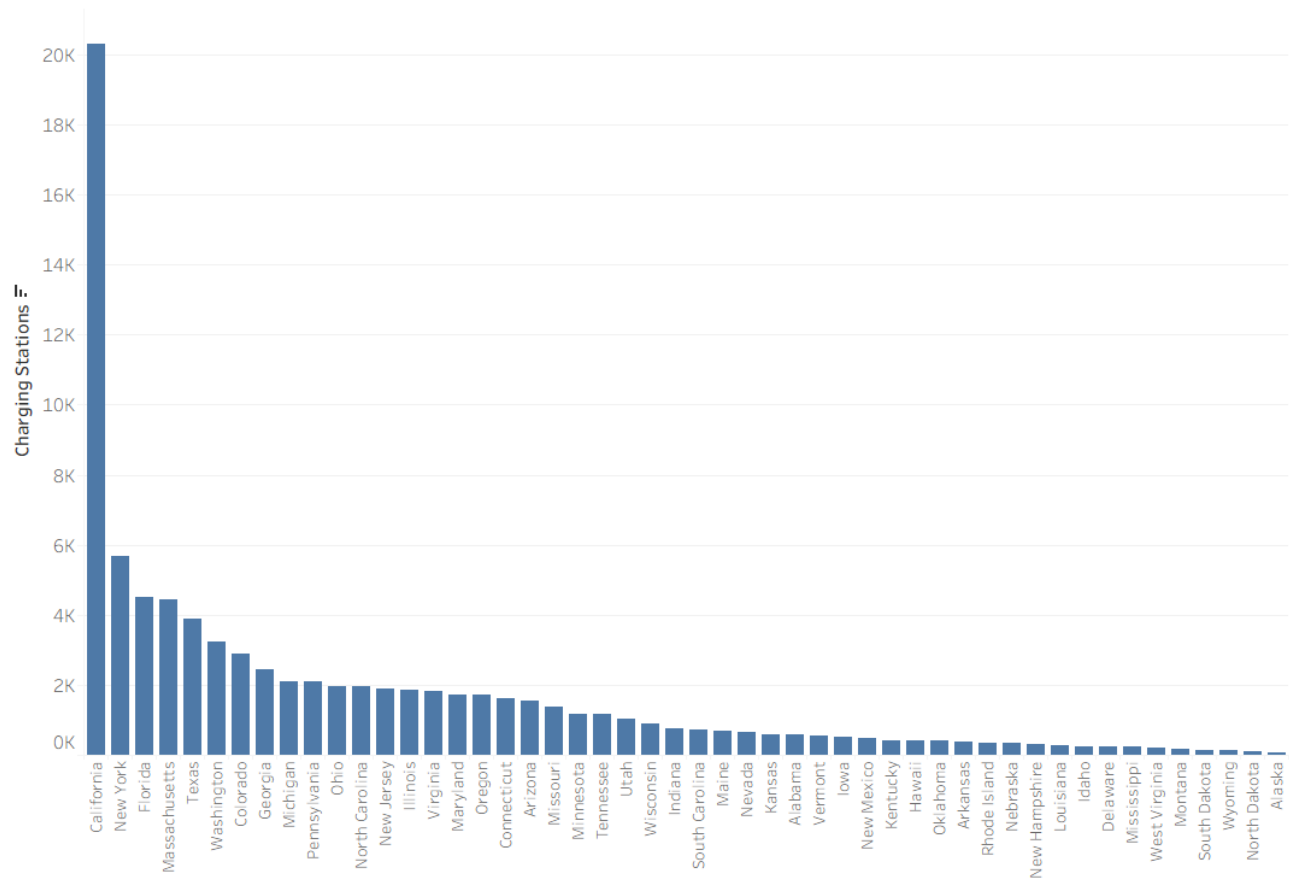
Hybrid/EV Adoption by State 2024 (Population Adjusted)



The map shows California having the largest H/EV-registration-to-population ratio by far, at nearly 10%. Other states with high proportions are Oregon, Washington, Colorado, and Vermont. Texas and New York have noticeably smaller proportions despite their large populations. The states with the smallest proportions are both in the Deep South: Louisiana and Mississippi, both with a proportion of about 1.5%.

The second visualization shows a bar graph of in-operation EV Charging stations per state. The x-axis corresponds to the fifty states, and the y-axis shows the number of charging stations in each state. The graph is sorted from greatest on the far left to least on the far right. This plot is the one that will receive up-to-date data from the NLR API going forward. Whenever NLR reports that a new electric charging station has opened or an old one has closed, it should be reflected here.

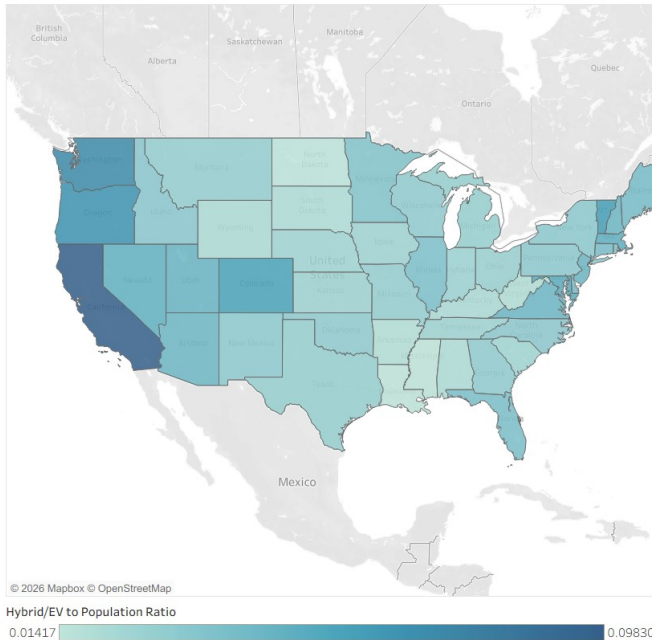
EV Stations per State (Updated Daily)



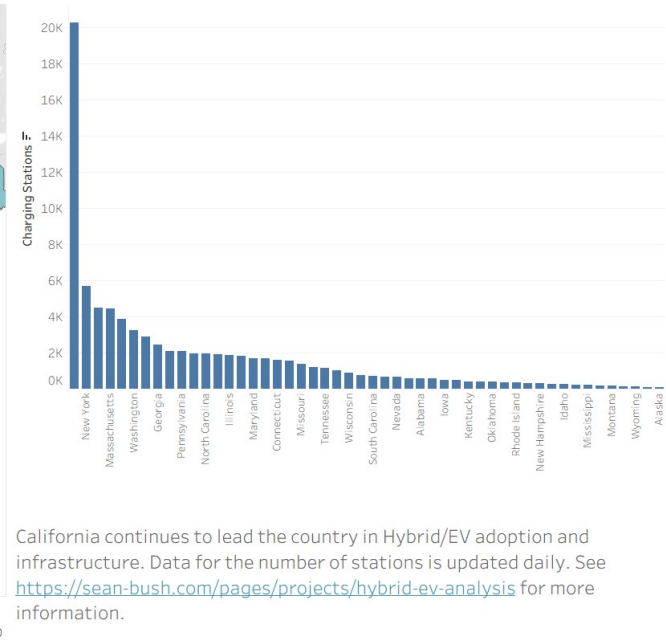
Unsurprisingly given the findings from the previous visualization, California has a significant lead over other states. As of the writing of this report, New York, Florida, and Texas are close behind. Interestingly, despite its relatively small population, Massachusetts has the fourth largest number of stations in the U.S.

The final assembled dashboard is as follows:

Hybrid/EV Adoption by State 2024 (Population Adjusted)



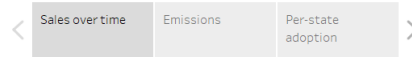
EV Stations per State (Updated Daily)



This dashboard specifically allows for filtering by state. As such, if one wanted to specifically analyze the Midwest, they could select the relevant states, and the bar graph would reflect those changes, making it easier to directly compare states that might have been originally far from each other in the graph and difficult to compare.

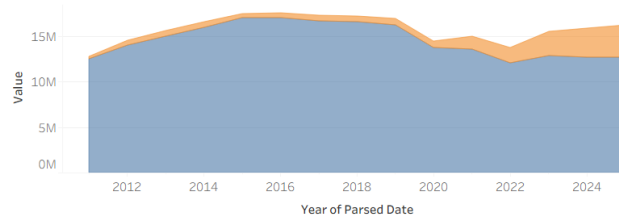
As stated earlier, the three dashboards we have discussed in this section are combined into one data story at the end of the workbook, and that story has been uploaded to the Tableau Public platform, where anyone is free to look through it.

U.S. Hybrid/EV Adoption Story

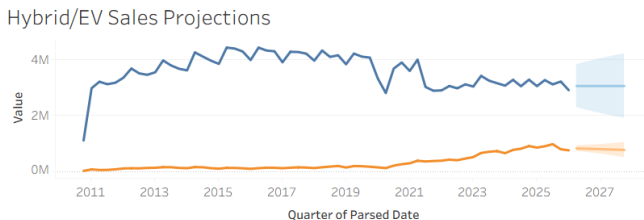


The State of Hybrid/EV Sales in the United States

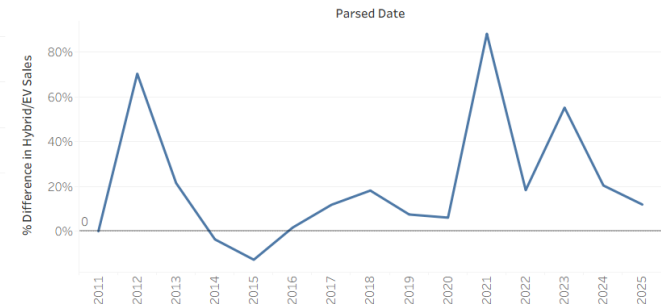
U.S. Car Sales by Year



Hybrid/EV Sales Projections



Hybrid/EV Percent Difference Per Year



While still maintaining a minority share of the market, Hybrid and EV vehicle sales continue to increase every single year. However, the market is not projected to change much in 2027.

tableau public

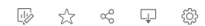
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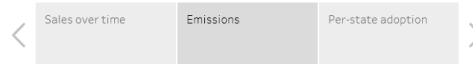
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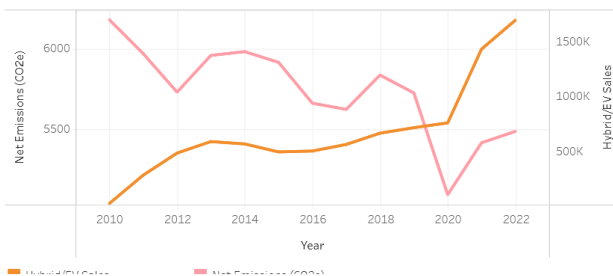
The State of Hybrid/Electric Vehicle Adoption in the U.S. by [Sean Bush](#) He/Him



U.S. Hybrid/EV Adoption Story

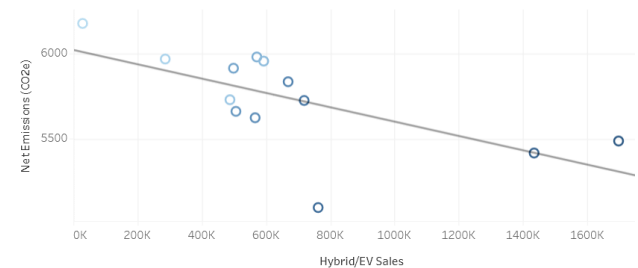


Net Emissions and Hybrid/EV Sales per Year



Correlation: Hybrid/EV Sales and Net

Net Emissions by Hybrid/EV Sales



A decrease in Net U.S. Emissions is correlated with an increase in Hybrid/EV Sales over time. This is one situation in which it is

Results

This project sought to create visualizations that would tell clear stories to answer four research questions:

- How have U.S. H/EV Sales been growing over the past several years?
- In the U.S., are emissions being reduced in concert with growing H/EV sales?
- How does H/EV adoption vary across states?
- How is adoption predicted to change over the next year?

The first question, how sales have been growing over the years, was addressed by the first dashboard. H/EV sales have consistently been growing, with only two years of the past fifteen where sales declined from the previous year. Over the time shown in this report, the portion of the market occupied by H/EVs has increased, though it still holds a minority share compared to gas-powered vehicles. It also accounts for most of the growth in the car market since 2020, as gas vehicle sales alone have stayed fairly constant for the last several years.

The first dashboard also addressed the fourth research question: how adoption is predicted to change. The fairly simple model predicted a slight decline in sales, but with a margin of error that would allow for a slight increase or a more significant decrease. Given that result and the general upward trend from the previous fifteen years, it is unclear how much growth can be expected in the immediate future, but immediate changes seem unlikely to be drastic.

The second dashboard addressed whether emissions are being reduced in concert with growing H/EV sales. There was a detectable negative correlation indicating that the variables are moving in time with each other, but the correlation was not significant enough to reveal a novel connection. There were far too many outside factors present to even begin to assess causality. Also, the correlation was not significant enough to track whether quick changes, say for a year or two, in emissions closely track with H/EV sales. It seems more likely that they are simply trending in opposite directions over time, and that any relationship between them was too minor to be detected with this simplistic analysis. This is expected, as visualization and simple analysis were the intended scope of this report, but it does suggest a direction for further analysis with more robust machine learning models.

The final dashboard addressed how H/EV adoption varied across states. It told a fairly consistent story: California leads other states by a significant margin. The other West Coast states also have a higher ratio of H/EV registrations to population than other states. As of the time of writing, states with larger populations such as California, New York, and Texas tend to lead with respect to the number of EV stations in operation. However, California still leads by a significant margin, and Massachusetts inexplicably has a very high number of stations. Further analysis might consider reporting these numbers also adjusted for inflation to be able to compare states not only on the number of EV stations within their borders, but the number of stations a resident of that state might reasonably be able to access. Though this is purely speculation and not a finding of this report, such an analysis would likely put states like Massachusetts much higher and states like Florida and Texas lower on the infrastructure readiness scale than the current bar graph suggests.

References

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